Seminar Question 7

"Explain in more detail the order-one uncertainty factor p in the estimate of $\Delta E_{\rm SO}$ in Eq. (16)."

From: A. Caputo *et al.*, *On the sensitivity of nuclear clocks to new physics*,
Phys. Rev. C (Letter), DOI 10.1103/l29n-gt5j; arXiv:2407.17526 (v1 long version, v2 = published)

A slow, self-contained walk from the nuclear-physics basics to the answer, with the missing steps reconstructed and checked numerically.

The short answer (read this first, then again at the end)

p in one paragraph

In the halo model, the *only* electromagnetic difference between the ground state and the isomer is the spin–orbit energy of one neutron moving in the Coulomb field of the ^{228}Th core. That energy is an integral of the neutron's radial probability density $|u(r)|^2$ against the weight $\frac{1}{r}\frac{dV_{\rm C}}{dr}$. Because the core has nearly uniform charge density, this weight is a *constant* inside the core (Ze^2/R_0^3) and falls off as Ze^2/r^3 outside. Pull the constant out of the integral and what remains is, essentially, the probability of finding the neutron *inside* the core (plus a small, suppressed tail piece). **That dimensionless number is p.** It is "order one" because the neutron is bound by 5.2 MeV and so lives mostly inside; it is *uncertain* because the paper does not know the neutron's wave function inside the core (only its universal tail outside), so it replaces the interior by "tail formula continued inward down to a cutoff ℓ". In that scheme $p \simeq 1$ and almost independent of ℓ; a more realistic interior wave function (we try a Woods–Saxon potential) gives $p \approx 0.66$. Either way $\Delta E_{\rm SO} \sim 100$ keV and the enhancement $K = \Delta E_{\rm SO}/\Delta E \sim 10^4$: p cannot make the "worst case" $K \to 0$ happen.

> UV length-scale, the charge density can be approximated by a Woods–Saxon form [66], see Eq. (5). The universal property $\rho \approx$ constant leads to $dV_{\rm C}/dr \propto r$, which conveniently cancels the UV sensitivity of the integral in Eq. (14); we then assume a constant $A(r)$, constrained by the normalization condition on $u(r)$, and numerically find
>
> $$\Delta E_{\rm SO} = E_{\rm SO}|_{j=5/2} - E_{\rm SO}|_{j=3/2} \approx 144\,p\,\text{keV}\,, \qquad (16)$$
>
> where the order-one factor p is a function of A and l, the latter being a short distance cut-off we introduced to regularize the integrals for $r \to 0$ (we find that the integral is almost independent of l for values $l \leq 4$ fm). A more complete nuclear model would predict a specific form for $A(r)$, eliminating the need for a regulator, and thereby determining p. Intuitively, p is related to the probability of finding the halo neutron inside the nucleus. The details of the computation are given in the supplementary material [63].

Figure 1: The passage under discussion (published version, p. 5, reproduced for study purposes). Everything below is an unpacking of these eleven lines.

Part 1 Where we are in the paper

Goal of this part

Remind yourself why a 100 keV number is being computed at all, and what role Eq. (16) plays.

The paper's logic in three steps:

1. The 8.36 eV transition energy of ^{229}Th is a near-cancellation, $\Delta E = \Delta E_{\rm em} + \Delta E_{\rm nuc}$ [their Eq. (1)], with each piece individually far larger than 8 eV.
2. If new physics jiggles $\alpha_{\rm em}$, the clock frequency shifts by an *enhanced* fraction $K = |\Delta E_{\rm em}|/\Delta E$ [Eq. (3)]. So one must estimate $\Delta E_{\rm em}$, the Coulomb-energy *difference* between the two nuclear states.
3. Two models are used. The geometric (deformed liquid drop) model gives $K \sim 10^4$–10^5 but allows $K \approx 0$ for fine-tuned shape parameters. The *halo model* — ^{228}Th core plus one $l = 2$ neutron — is the second, and Eq. (16) is its final result:

$$\Delta E_{\rm SO} = E_{\rm SO}\big|_{j=5/2} - E_{\rm SO}\big|_{j=3/2} \approx 144\,p\,\text{keV}. \tag{16}$$

Why the halo model produces only one term

In the halo picture the ^{228}Th core is identical in the ground state and the isomer; the two states differ only in how the last neutron's spin is aligned relative to its orbital motion ($j = 5/2$ vs. $j = 3/2$). A neutron is neutral, so the Coulomb energy of the core does not change at all. The single surviving electromagnetic difference is the interaction of the neutron's *magnetic moment* with the Coulomb field it moves through — the spin–orbit term. So $\Delta E_{\rm em} \to \Delta E_{\rm SO}$ exactly, and Eq. (16) *is* the halo-model prediction for $\Delta E_{\rm em}$.

Part 2 Nuclear-physics refresher (slow)

Goal of this part

Build, one piece at a time, every ingredient that appears in Eqs. (14)–(16): the core-plus-neutron picture, the labels l, s, j, why a neutral particle has a spin–orbit energy, the radial wave function $u(r)$ and its tail, and the Coulomb potential of a charged sphere.

2.1 A heavy nucleus as "core + one neutron"

^{229}Th has $Z = 90$ protons and $N = 139$ neutrons. ^{228}Th ($N = 138$) is *even–even*: all nucleons pair off into spin-0 couples, so its ground state has $j^\pi = 0^+$ and carries no angular momentum. Adding one more neutron gives ^{229}Th, and in the simplest (shell-model / halo) picture *that single neutron carries all the angular momentum* of the nucleus. This is the nuclear analogue of an alkali atom: an inert closed core plus one active valence particle. The ground state $5/2^+$ and the 8 eV isomer $3/2^+$ are then two states of *the same neutron orbit*, differing only in spin orientation (Fig. 2).

2.2 The labels l, s, j and the quantity $\boldsymbol{L}\cdot\boldsymbol{S}$

The neutron has orbital angular momentum $l = 2$ (spectroscopic letter "d", hence "d-wave halo") and spin $s = \frac{1}{2}$. They add to a total $j = l \pm \frac{1}{2}$, i.e. $j = 5/2$ or $3/2$. Any interaction of the form $f(r)\,\boldsymbol{L}\cdot\boldsymbol{S}$ (spin–orbit) is diagonal in j and takes the value

$$\langle \boldsymbol{L}\cdot\boldsymbol{S}\rangle = \tfrac{1}{2}\big[j(j+1) - l(l+1) - s(s+1)\big]\hbar^2 = \tfrac{1}{2}\big[j(j+1) - l(l+1) - \tfrac{3}{4}\big]\hbar^2,$$

which is exactly the square bracket in the paper's Eq. (14). Numbers for $l = 2$:

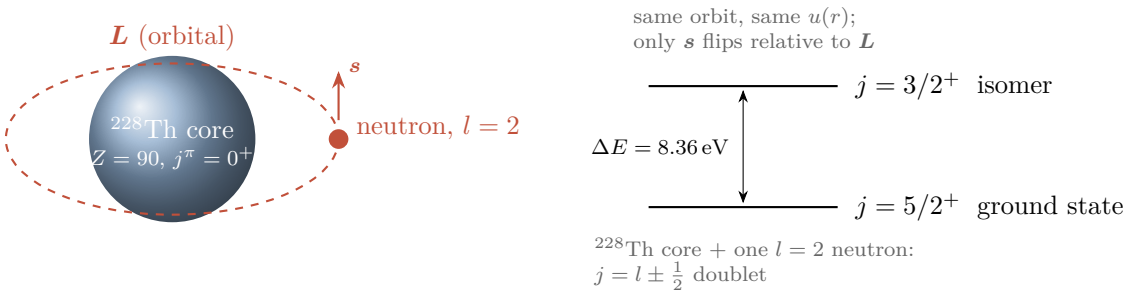


Figure 2: The halo picture. Left: an inert, spherical, charged core and one neutron in a d-wave ($l = 2$) orbit with spin $\boldsymbol{s}$. Right: the two members of the $j = l \pm \frac{1}{2}$ doublet are the ground state and the isomer. Spin–orbit physics decides their tiny splitting.

	$j(j+1) - l(l+1) - \frac{3}{4}$	$\langle \boldsymbol{L}\cdot\boldsymbol{S}\rangle/\hbar^2$	state
$j = l + \frac{1}{2} = 5/2$	$\frac{35}{4} - 6 - \frac{3}{4} = \;\;2$	$+1$	ground
$j = l - \frac{1}{2} = 3/2$	$\frac{15}{4} - 6 - \frac{3}{4} = -3$	$-\frac{3}{2}$	isomer
difference	**5**	$\frac{5}{2} = \frac{2l+1}{2}$	

So whatever the radial integral turns out to be, the *difference* between the two states is $5\times$ (radial integral) $\times$ (constants). The "5" is pure angular-momentum bookkeeping and carries no uncertainty at all.

2.3 Why a neutral neutron feels the Coulomb field

A neutron has no charge, but it has a magnetic moment $\boldsymbol{\mu} = \mu_n \frac{e\hbar}{2m_n}\boldsymbol{\sigma}$ with $\mu_n = -1.91$ (in nuclear magnetons). A magnetic moment moving with velocity $\boldsymbol{v}$ through an electric field $\boldsymbol{E}$ sees, in its own rest frame, a magnetic field $\boldsymbol{B}' \simeq -\boldsymbol{v}\times\boldsymbol{E}/c^2$, and pays an energy $-\boldsymbol{\mu}\cdot\boldsymbol{B}'$. For a spherical core the field is radial, $e\boldsymbol{E} = -\frac{dV_{\rm C}}{dr}\hat{\boldsymbol{r}}$, so $\boldsymbol{v}\times\boldsymbol{E} \propto \frac{1}{r}\frac{dV_{\rm C}}{dr}\,(\boldsymbol{r}\times\boldsymbol{p})/m_n \propto \frac{1}{r}\frac{dV_{\rm C}}{dr}\boldsymbol{L}$, and the energy becomes

$$H_{\rm SO} \;\propto\; \frac{\mu_n}{m_n^2}\,\frac{1}{r}\,\frac{dV_{\rm C}}{dr}\;\boldsymbol{L}\cdot\boldsymbol{S}.$$

Taking the expectation value in a state with radial wave function $u(r)$ gives the paper's

$$E_{\rm SO} = \frac{e\,\mu_n}{2m_n^2}\Big[j(j+1) - l(l+1) - \tfrac{3}{4}\Big]\int_0^\infty \frac{[u(r)]^2}{r}\,\frac{dV_{\rm C}}{dr}\,dr. \tag{14}$$

Here $V_{\rm C}(r)$ is the Coulomb potential energy *a proton* would have at radius r (the neutron uses it only through its gradient, i.e. the electric field), and $u(r)$ is normalised, $\int_0^\infty u^2\,dr = 1$.

Atomic fine structure, but with a twist

This is the same mechanism that splits the sodium D lines: an electron's spin feels the motional magnetic field of the nucleus. Two differences: (i) the "charge" driving the coupling is the neutron's *anomalous* magnetic moment μ_n rather than an electric charge, and (ii) the field source is an extended charged sphere, not a point — which, as we will see, is the whole reason p is well behaved.

2.4 The radial wave function $u(r)$: a universal tail and an unknown interior

Outside the core the neutron feels no nuclear force, so its wave function is the free solution of the Schrödinger equation at negative energy $-E_B$, with $E_B = 5.2\,\text{MeV}$ the neutron separation energy.

For $l = 2$ this is

$$u(r) = A(r)\, e^{-\gamma r}\Big(1 + \frac{3}{\gamma r} + \frac{3}{(\gamma r)^2}\Big), \qquad \gamma = \frac{\sqrt{2 m_n E_B}}{\hbar} = 0.50\,\mathrm{fm}^{-1}, \quad \gamma^{-1} = 2.0\,\mathrm{fm}. \tag{15}$$

Three things to absorb:

- **The tail is universal.** Given E_B and l, the *shape* at large r is fixed; only its overall size A — the *asymptotic normalisation constant* (ANC) — depends on what happens inside. That is why the paper writes $A(r) \to A$ (constant) at large r.
- **The formula is wrong at small r.** As $r \to 0$, $u \to 3A/(\gamma r)^2$: it blows up. A real bound-state wave function must vanish at the origin like $r^{l+1} = r^3$ (the centrifugal barrier keeps the neutron away from $r = 0$). The paper acknowledges this by letting A become a function $A(r)$ inside the core, which they do not know.
- **Two fermi is not very "halo".** $\gamma^{-1} = 2.0\,\mathrm{fm}$ is smaller than the core radius $\sim$ 6–7 fm: the tail is short, and the neutron spends most of its time inside. The authors say as much ("halo EFT should only be considered a model here"). Keep this in mind — it is the physical reason p will come out close to 1.

2.5 The Coulomb potential of a uniformly charged sphere

Nuclear charge densities are nearly flat inside and drop at the surface (Woods–Saxon). Approximating the core as a uniform sphere of radius R_0 gives the familiar

$$V_{\mathrm{C}}(r) = \begin{cases} \dfrac{Ze^2}{2R_0}\Big(3 - \dfrac{r^2}{R_0^2}\Big), & r < R_0 \\[2ex] \dfrac{Ze^2}{r}, & r > R_0 \end{cases} \qquad \Longrightarrow \qquad \frac{1}{r}\frac{dV_{\mathrm{C}}}{dr} = \begin{cases} -Ze^2/R_0^3 & r < R_0 \\ -Ze^2/r^3 & r > R_0. \end{cases}$$

(The overall sign only fixes which of the two j states lies higher; we track magnitudes.) Figure 3 shows both functions. The right-hand panel is the crucial one: the weight that multiplies $|u|^2$ in Eq. (14) is *flat* inside the core.

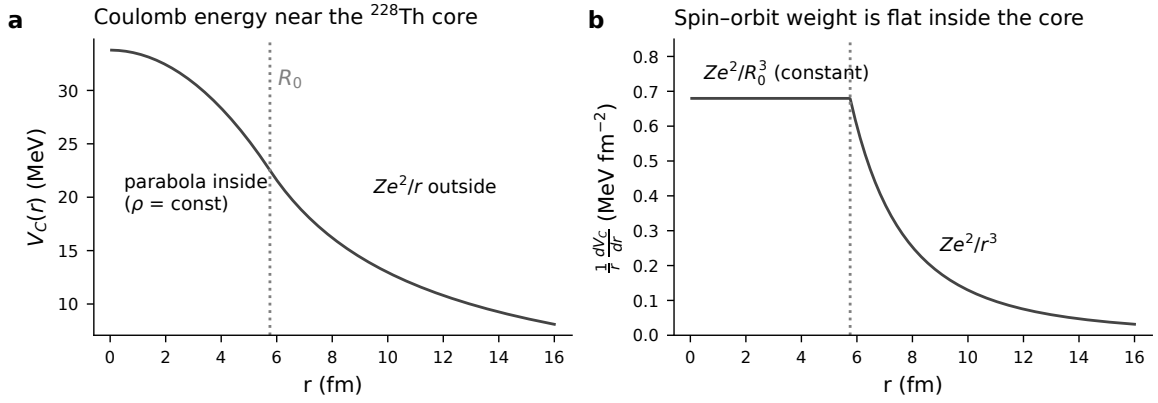


Figure 3: Coulomb potential energy of a proton (**a**) and the spin–orbit weight $\frac{1}{r}\frac{dV_{\mathrm{C}}}{dr}$ (**b**) for a uniformly charged sphere with $Z = 90$ and $R_0 = 5.76\,\mathrm{fm}$. Inside the core the weight is the constant Ze^2/R_0^3; outside it falls as Ze^2/r^3.

For an experimental HEP reader

A point-charge core would give $\frac{1}{r}\frac{dV_{\mathrm{C}}}{dr} = Ze^2/r^3$ everywhere, and the integral in Eq. (14) would be dominated by — and divergent at — $r \to 0$, i.e. by physics the model knows nothing about.

The extended charge distribution acts like a physical UV regulator: this is what the paper means by "$\rho \approx$ constant leads to $dV_{\rm C}/dr \propto r$, which conveniently cancels the UV sensitivity". Think of it as an IR-safe observable: the result is controlled by the bulk, not by the unresolved short-distance structure.

Part 3 Dissecting Eq. (14)–(16): where p is born

Goal of this part

Do the algebra the paper skips. Show that the radial integral factorises into (known constant) $\times\, p$, identify p as an inside-the-core probability, reproduce the number 144, and understand why a cutoff ℓ enters and why the result barely depends on it.

3.1 Splitting the integral at the core surface

Insert the piecewise weight into Eq. (14). The integral becomes (v1 of the preprint, Eq. (22))

$$\int_0^\infty \frac{u^2}{r}\frac{dV_{\rm C}}{dr}\,dr = \frac{Ze^2}{R_0^3}\left[\underbrace{\int_0^{R_0} u^2\,dr}_{P_{\rm in}} + \underbrace{R_0^3\int_{R_0}^\infty \frac{u^2}{r^3}\,dr}_{\text{tail term}}\right] \equiv \frac{Ze^2}{R_0^3}\,p.$$

This bracket is p

- $P_{\rm in} = \int_0^{R_0} u^2 dr$ is literally the *probability that the neutron is inside the charged core.* This is the paper's "intuitively, p is related to the probability of finding the halo neutron inside the nucleus".
- The tail term is the outside probability, but down-weighted by $(R_0/r)^3 \le 1$. Since $P_{\rm in} + P_{\rm out} = 1$, we get the bound $\boxed{p \le 1}$ (for a properly normalised u).
- p is dimensionless and depends *only* on the shape of $|u(r)|^2$ relative to R_0 — i.e. on the interior wave function $A(r)$ and on whatever regulator is used to stand in for it. That is exactly the paper's "p is a function of A and ℓ".

3.2 Reproducing the 144

Combine the angular factor (5), the constants, and p:

$$\Delta E_{\rm SO} = \underbrace{\frac{|\mu_n|}{2}\left(\frac{\hbar c}{m_n c^2}\right)^2 \frac{Ze^2}{R_0^3} \times 5}_{\text{all known}} \times\, p.$$

With $|\mu_n| = 1.913$, $\hbar c/m_n c^2 = 0.210\,\text{fm}$, $e^2 = \alpha\hbar c = 1.44\,\text{MeV fm}$, $Z = 90$:

choice of R_0	prefactor (keV)	comment
$\sqrt{\langle r^2\rangle} = 5.756\,\text{fm}$ (rms radius)	**143.4**	reproduces the paper's 144 with $p = 1$
$\sqrt{5/3}\,\sqrt{\langle r^2\rangle} = 7.43\,\text{fm}$ (equivalent sphere)	66.6	the "textbook" uniform-sphere radius
$1.2\,A^{1/3} = 7.34\,\text{fm}$	69.1	standard nuclear-radius rule

So the number 144 corresponds to setting the sphere radius equal to the measured rms charge radius and p to one. (Our reconstruction; the supplementary material is not in the arXiv PDF.)

Two small caveats about the "144"

(i) R_0 convention. For a uniform sphere the rms radius is $\sqrt{3/5}\,R_0$, so the self-consistent R_0 is 7.4 fm, which would give $\approx 67\,\text{keV} \times p$. Because $\Delta E_{\rm SO} \propto R_0^{-3}$, a 30% change in R_0 is a factor 2.2 in the prefactor. This ambiguity is of the same size as the uncertainty in p itself, and you may reasonably regard it as *part* of the order-one factor. It does not change the conclusion $\Delta E_{\rm SO} \sim 10^2$ keV.
(ii) Version difference. arXiv v1, Eq. (19), carries an extra factor $\frac{1}{2}$ in front of the square bracket relative to the published Eq. (14); both versions quote $144\,p$ keV. Our numbers reproduce 144 with the published form, so treat v1's $\frac{1}{2}$ as a typo.

3.3 Why a cutoff ℓ at all, and why the answer does not care

The paper does not know $A(r)$ inside the core, so it makes the crudest possible guess: keep A constant and use the tail formula (15) all the way in. But (15) diverges as $1/r^2$, so $\int u^2 dr \sim \int dr/r^4$ is infinite at $r \to 0$. Hence a *regulator*: start all integrals at $r = \ell$, then fix A from the normalisation $\int_\ell^\infty u^2 dr = 1$. In this scheme

$$p(\ell) = \frac{\int_\ell^{R_0} u^2 dr + R_0^3 \int_{R_0}^\infty u^2 r^{-3} dr}{\int_\ell^\infty u^2 dr}.$$

We evaluated this (Fig. 5, blue): $p = 1.000, 0.999, 0.993, 0.970$ for $\ell = 1, 2, 3, 4$ fm — flat, exactly the "almost independent of ℓ for $\ell \le 4$ fm" stated in the paper. The reason is simple once you see Fig. 4a: the singular tail formula piles essentially all of the (normalised) probability just above the cutoff, and as long as $\ell < R_0$ that pile sits *inside* the core, so $P_{\rm in} \to 1$. Only when ℓ approaches R_0 does probability leak outside and p drops.

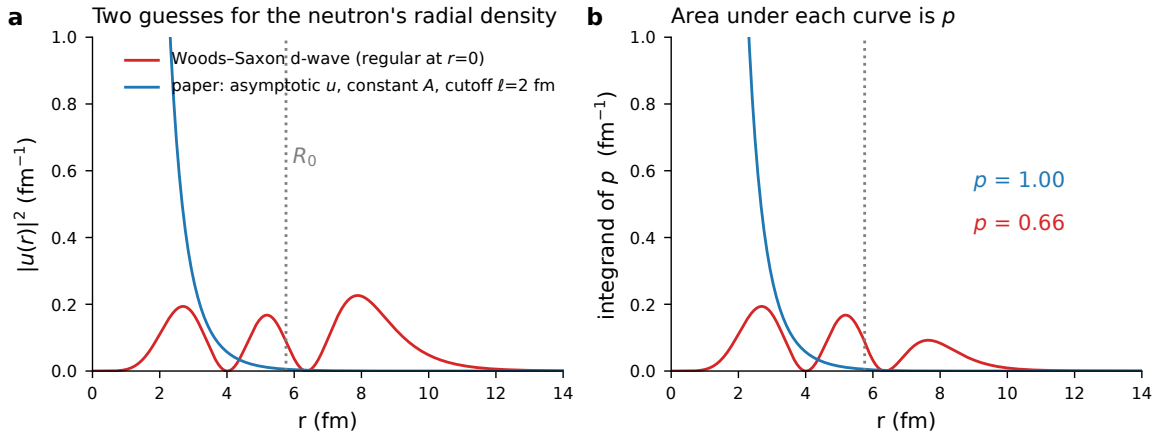


Figure 4: **a** Two candidate radial densities $|u(r)|^2$ for the $l = 2$ neutron bound by 5.2 MeV: the paper's prescription (blue: asymptotic form, constant A, cutoff $\ell = 2$ fm) and a bound state in a Woods–Saxon well (red: $V_0 = 48$ MeV, $R = 1.27A^{1/3} = 7.76$ fm, $a = 0.67$ fm; regular at the origin, two radial nodes, tail matched to Eq. (15)). Both are normalised to 1. **b** The same densities multiplied by the (normalised) spin–orbit weight $R_0^3 \frac{1}{r}\frac{dV_{\rm C}}{dr}/Ze^2$; the area under each curve is p. Dotted line: $R_0 = 5.76$ fm.

Regulator independence is not the same as model independence

The flatness in ℓ tells you the *regulator* is harmless. It does not tell you that $p \simeq 1$ is right — it tells you that within the assumption "constant A down to short distances" the answer is $p \simeq 1$. The true uncertainty in p is the assumption, not the cutoff.

3.4 What "a more complete nuclear model" would do

To see how much the interior matters we solved the radial Schrödinger equation for an $l = 2$ neutron in a standard Woods–Saxon potential, adjusting the depth so that the binding energy is 5.2 MeV (depth 48 MeV, two radial nodes — the $3d$ orbital expected near $N = 139$), matched to the tail of Eq. (15) at large r and normalised. This wave function needs *no* regulator; it is regular at the origin and oscillates inside the core (Fig. 4, red). The result is

$$p_{\mathrm{WS}} = 0.66, \qquad P_{\mathrm{in}}(r < 5.76\,\mathrm{fm}) = 0.46, \qquad \text{ANC } A = 19.5\,\mathrm{fm}^{-1/2},$$

to be compared with $p \simeq 1.0$ in the paper's scheme. Two lessons:

- A realistic interior spreads the neutron over the whole core and pushes a third of the probability past $R_0 = 5.76$ fm, where the weight $\propto r^{-3}$ is smaller. p drops — but only to $\sim 2/3$, still "order one".
- This is an illustration, not a prediction. ^{229}Th is actually deformed (Nilsson label $[633]\frac{5}{2}$ for the ground state), the neutron orbit is not a pure d-wave, and the single-particle strength is shared (spectroscopic factor < 1). A serious model (density functional, or a shell model in a deformed basis) would predict $A(r)$ and hence p, which is what the paper calls "eliminating the need for a regulator and thereby determining p".

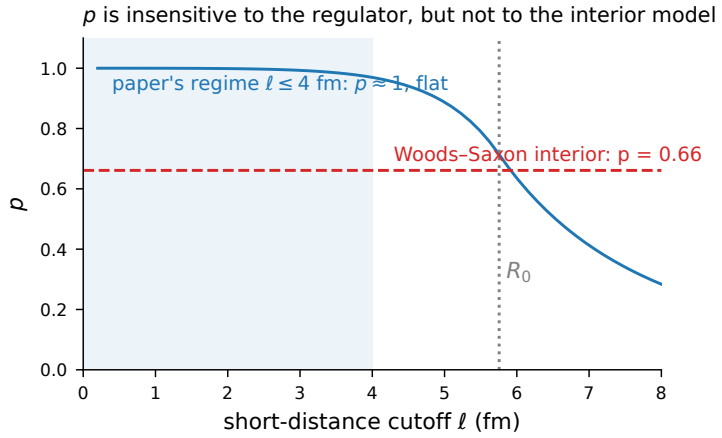


Figure 5: p in the paper's constant-A scheme as a function of the short-distance cutoff ℓ (blue; shaded: the paper's quoted range $\ell \le 4$ fm), compared with the regulator-free Woods–Saxon value (red dashed). $R_0 = 5.76$ fm throughout.

Part 4 So what, precisely, is uncertain about p?

Goal of this part

Separate the things that live inside p from things the paper treats separately, so you can answer follow-up questions in the seminar.

Source of uncertainty	What it does	Size / status
Interior wave function $A(r)$	Sets how much of $\lvert u\rvert^2$ sits inside R_0; the paper's constant-A guess gives $p \simeq 1$, a Woods–Saxon interior gives 0.66	*The* content of p; factor ~ 1.5
Cutoff ℓ	Regulates the $1/r^4$ divergence of the tail formula	$< 3\%$ for $\ell \le 4\,$fm (paper); negligible
Sphere radius R_0	Prefactor $\propto R_0^{-3}$; rms radius vs. equivalent sphere	factor 2.2; arguably part of p
Not inside p, but relevant to the whole estimate:		
Same $u(r)$ for $j = 5/2$ and $3/2$	Spin–orbit treated as a small perturbation; separation energies nearly equal	assumed; testable (see Part 5)
Same ANC A for both states	Enters normalisation; if different, $\Delta E_{\rm SO}$ changes	assumed; $\sigma_{\rm cap}^{\rm iso} = \frac{2}{3}\sigma_{\rm cap}^{\rm gs}$ test
In-medium neutron magnetic moment	Overall scale $\propto \mu_n$	the $\sim 50\%$ miss in $\Delta\langle r^2\rangle$ hints at this
Higher-order / many-body corrections	Core polarisation, exchange terms	not estimated

For an experimental HEP reader

Several analogies may help you file p mentally:

- **Acceptance × efficiency.** $\Delta E_{\rm SO} = (\text{known rate}) \times p$, with p the fraction of the "signal" (the neutron) that lands in the "fiducial volume" (the core), weighted by a known response (R_0^3/r^3 outside). You would never quote such a factor to better than its modelling allows, but you also know it cannot be 10^{-4}.
- **Regulator vs. scheme.** The cutoff ℓ is like a renormalisation scale: the physical result should not depend on it, and here it does not. The constant-A assumption is like choosing a parton-shower model: that is where the real systematic lives.
- **Order-one factor ≈ K-factor.** A leading-order estimate with an unknown but $\mathcal{O}(1)$ multiplicative correction, expected to be refined by a better calculation.
- **ANC ≈ spectroscopic factor.** In transfer-reaction language, A^2 is what a (d,p) or (n,γ) measurement constrains; it normalises the single-particle tail.

Part 5 Why it matters, and how to measure p

Goal of this part

Connect p back to the paper's headline: the enhancement factor K and the (im)possibility of the worst case.

scenario	p	$\Delta E_{\rm SO}$ (keV)	$K = \Delta E_{\rm SO}/\Delta E$
paper's scheme (constant A, $\ell \le 4\,$fm)	1.0	143	1.7×10^4
Woods–Saxon interior (this guide)	0.66	95	1.1×10^4
equivalent-sphere $R_0 = 7.4\,$fm, $p = 1$	—	67	0.8×10^4
hypothetical: neutron 90% outside the core	~ 0.1	~ 14	$\sim 2 \times 10^3$

The point of the whole exercise

In the geometric model K could be tuned to zero by unlucky shape parameters. In the halo model $\Delta E_{\rm SO}$ is a single term with a *definite* sign and a magnitude $\propto p$, and p is a probability-like quantity for a neutron bound by 5.2 MeV with a 2 fm tail. Making $K \lesssim 1$ would require $p \lesssim 10^{-4}$, i.e. a neutron that essentially never enters the core — incompatible with its binding

energy. That is why the paper can say "within the halo model, the worst case scenario is never realized, and $K_{\rm em} \gtrsim 10^4$." The uncertainty in p is a factor of a few on K, not a threat to the enhancement.

How would one pin p down experimentally? The paper's outlook lists the handles:

- **Neutron capture on ^{228}Th.** The d-wave capture amplitude is proportional to the ANC A, so $\sigma_{\rm cap}$ constrains the overlap of the neutron with the core — i.e. p.
- **Capture into ground state vs. isomer.** Halo EFT predicts $\sigma_{\rm cap}^{\rm iso} = \frac{2}{3}\sigma_{\rm cap}^{\rm gs}$; a deviation would signal different A (and different u) for the two states, undermining the "same $u(r)$" assumption.
- **Transfer reactions** such as ^{228}Th$(d,p)^{229}$Th measure spectroscopic factors for the $5/2^+$ and $3/2^+$ states directly.
- **Better theory:** a nuclear-structure calculation that outputs $A(r)$ removes the need for a regulator and *predicts* p.

Key numbers

quantity	value	where from
ΔE (isomer energy)	8.355 733 554 021(8) eV	paper, Ref. [40]
E_B (neutron separation energy of ^{229}Th)	5.2 MeV	paper, Eq. (15)
$\gamma = \sqrt{2m_n E_B}/\hbar$; γ^{-1}	0.50 fm^{-1}; 2.0 fm	computed
μ_n	$-1.913\,\mu_N$	paper, Eq. (10)
$\sqrt{\langle r^2\rangle}$ of ^{229}Th	5.7557(143) fm	paper, Ref. [56,59]
angular factor difference, $l = 2$	5	$[j(j+1) - l(l+1) - \frac{3}{4}]_{5/2} - [\dots]_{3/2}$
prefactor with $R_0 = 5.756$ fm	143.4 keV	computed; matches "144"
prefactor with $R_0 = 7.43$ fm	66.6 keV	computed
$p(\ell)$, constant-A scheme, $\ell = 1,2,3,4$ fm	1.000, 0.999, 0.993, 0.970	computed
p, Woods–Saxon interior	0.66	computed (illustration)
K for $p = 1$ / $p = 0.66$	1.7×10^4 / 1.1×10^4	$\Delta E_{\rm SO}/\Delta E$

Glossary

Halo nucleus
A core plus one (or two) weakly bound nucleons whose wave function extends well outside the core. ^{229}Th is a marginal case ($\gamma^{-1} = 2$ fm).

d-wave
Orbital angular momentum $l = 2$. The centrifugal barrier $l(l+1)/r^2$ makes the wave function vanish as r^3 at the origin and makes short-distance physics matter more than for s-waves.

Spin–orbit doublet
The pair $j = l \pm \frac{1}{2}$ from one orbit; here $5/2^+$ (gs) and $3/2^+$ (isomer).

Asymptotic normalisation constant (ANC), A
The coefficient of the universal exponential tail of a bound-state wave function; encodes the interior in a single number.

UV / short-distance sensitivity
Dependence of an integral on the region $r \to 0$ where the model is not trusted. The uniform-sphere Coulomb field removes it here.

Regulator / cutoff ℓ
A lower limit on the radial integrals that stands in for the unknown interior; results should not depend on it, and they do not.

Woods–Saxon
The standard smooth-edged nuclear density or potential, $[1 + e^{(r-R)/a}]^{-1}$, with radius $R \approx 1.2A^{1/3}$ fm and diffuseness $a \approx 0.5$–0.7 fm.

Enhancement factor K
$|\Delta E_{\mathrm{em}}|/\Delta E$: how much more sensitive the nuclear clock is to a change in α_{em} than the bare transition energy would suggest.

Discussion questions for the seminar

1. Show that $p \leq 1$ follows from the normalisation of $u(r)$. Under what modification of the charge density (e.g. a soft skin, $z \neq 0$) could the bound be violated, and by how much?
2. The paper says the integral is "almost independent of ℓ for $\ell \leq 4$ fm". Using Fig. 4a, explain in words why that is true *and* why it does not prove $p \simeq 1$.
3. The prefactor scales as R_0^{-3}. If the authors had used the equivalent-sphere radius (7.4 fm) instead of the rms radius, the headline would read "$\approx 67\,p$ keV". Does the paper's conclusion change? Should this ambiguity be absorbed into p?
4. Our Woods–Saxon interior gives $p = 0.66$. List three ways in which that toy model is still unrealistic for ^{229}Th (think: deformation, pairing, spectroscopic factors). Which direction would each push p?
5. Why does the same paper get a $\sim 50\%$ miss on $\Delta\langle r^2\rangle$ [Eq. (11)] but an almost exact match on ΔQ [Eqs. (12)–(13)], and how would an in-medium renormalisation of μ_n propagate into ΔE_{SO}?
6. Design the neutron-capture measurement that would constrain A: what target, what observable, what precision on σ_{cap} would translate into, say, 20% on p?

Suggested reading order in the original

1. Published version, p. 4, "Th-229 as a Halo Nucleus" up to Eq. (13): the model and its two successful checks.
2. Eqs. (14)–(15) and the paragraph on $A(r)$ and the ANC.
3. Top of p. 5 through Eq. (16) (the crop at the start of this guide).
4. arXiv v1 (long version), Eqs. (19)–(23): the uniform-sphere potential and the split integral that this guide fills in.
5. Summary and Outlook: the neutron-capture proposals that would fix A and hence p.

Numerical values in this guide were computed for the seminar from the paper's stated inputs (E_B, μ_n, Z, $\langle r^2\rangle$); the Woods–Saxon comparison is an illustration of interior-model dependence, not a result of the paper.